



Real Estate Analytics



AI Bootcamp by Quera

Project for Data Analysis and Modeling on Real Estate Ads



AI Bootcamp by Quera



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• Data Analysis and Modeling •



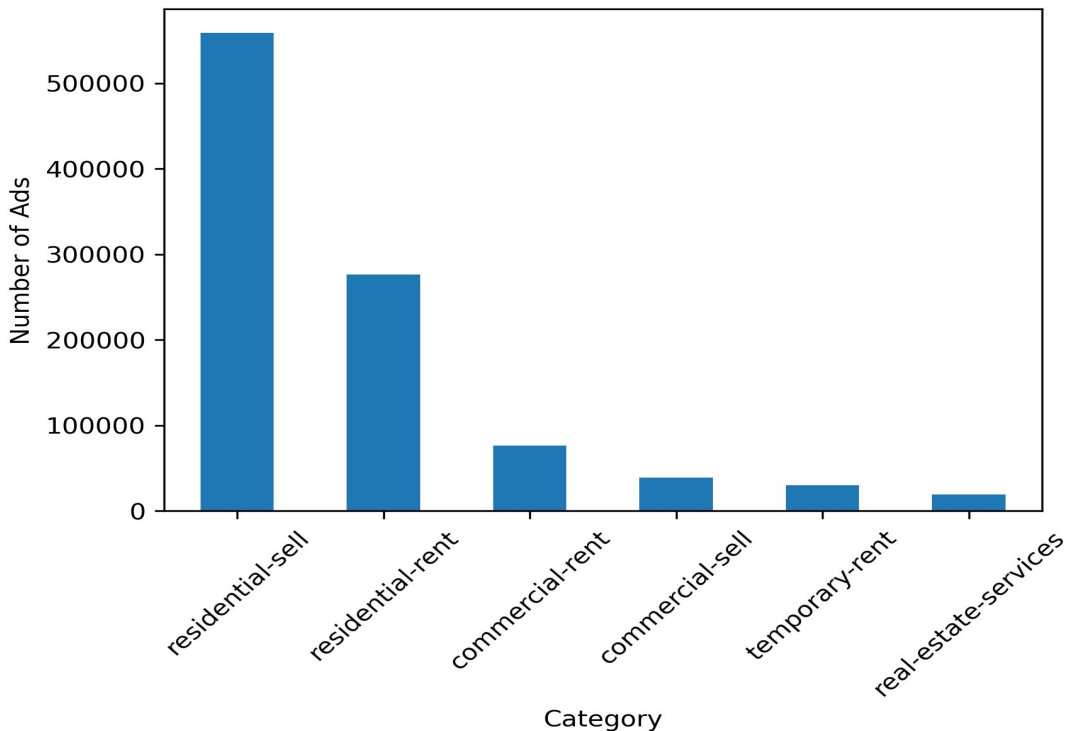
• Real Estate Ads •



Distribution of Ads by Category Tier 2

Conclusion

- Residential sell has the highest number of ads.
- Residential rent is the second largest category.
- Other categories have much fewer ads.

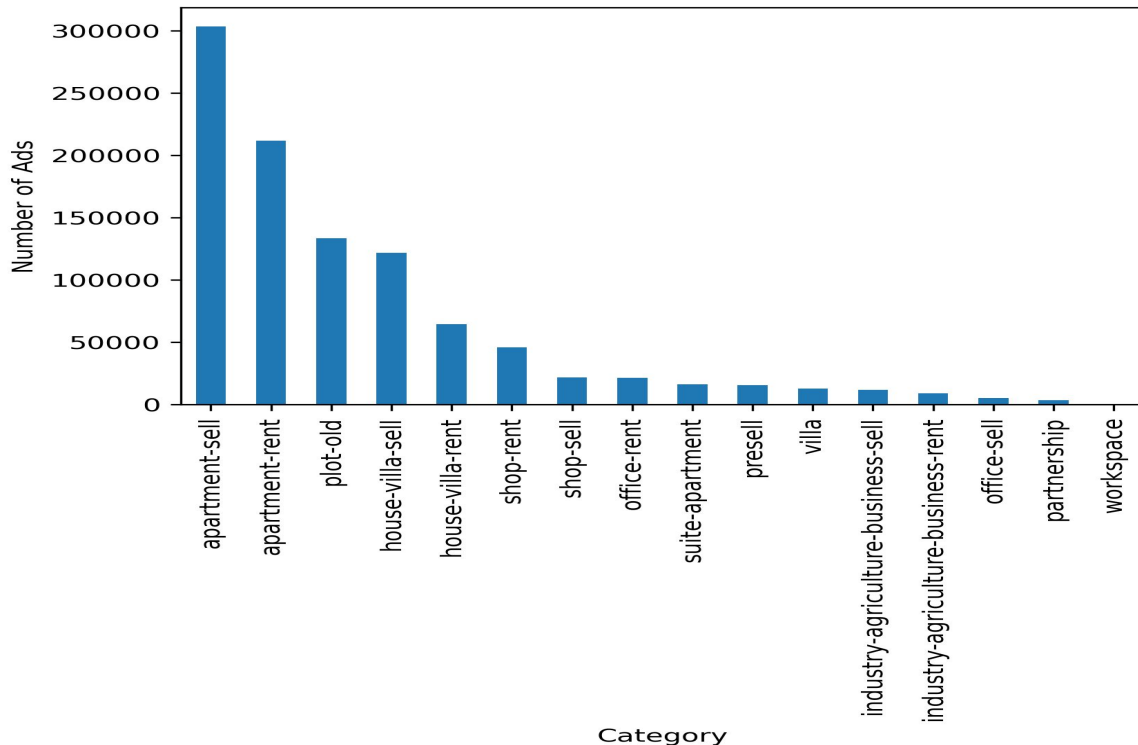




Distribution of Ads by Category Tier 3

Conclusion

- Apartment sell is the most common category.
- Apartment rent is the second largest category.
- House and plot listings are also common, while the remaining categories have relatively few ads.

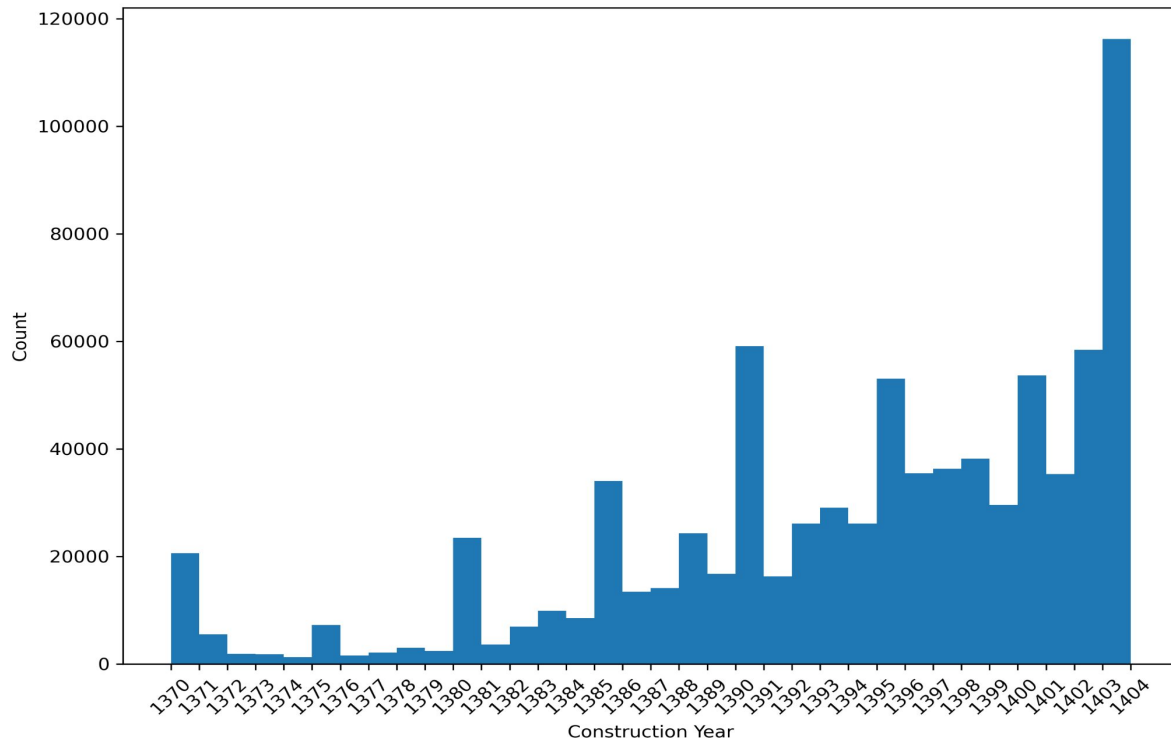




Distribution of Ads by Construction Year

Conclusion

- Most ads are for recently constructed properties.
- The number of ads generally increases with newer construction years.
- Older properties appear less frequently in the dataset.

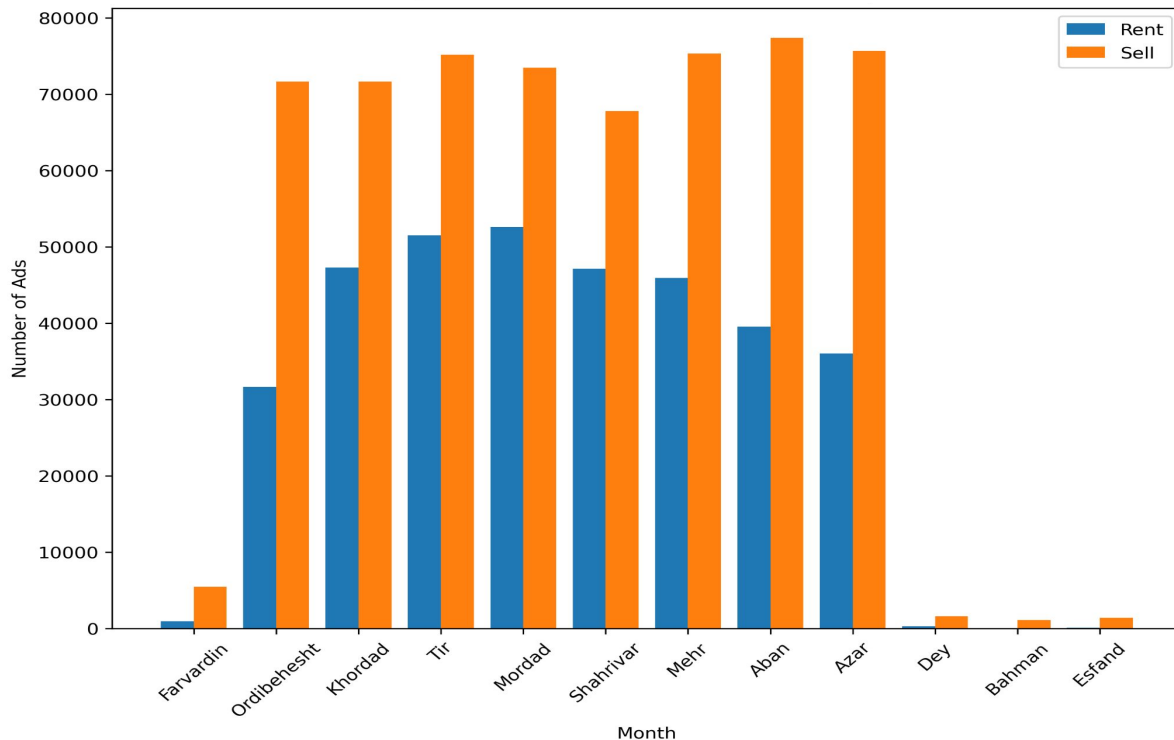




Distribution of Ads by Month(Rent vs Sell)

Conclusion

- Buy and rent listings are lower in Farvardin, Dey, Bahman, and Esfand.
- Winter weather makes moving and property visits more difficult.
- The end of the year and Nowruz holidays reduce housing market activity.
- Listings peak in summer due to school holidays and favorable weather.

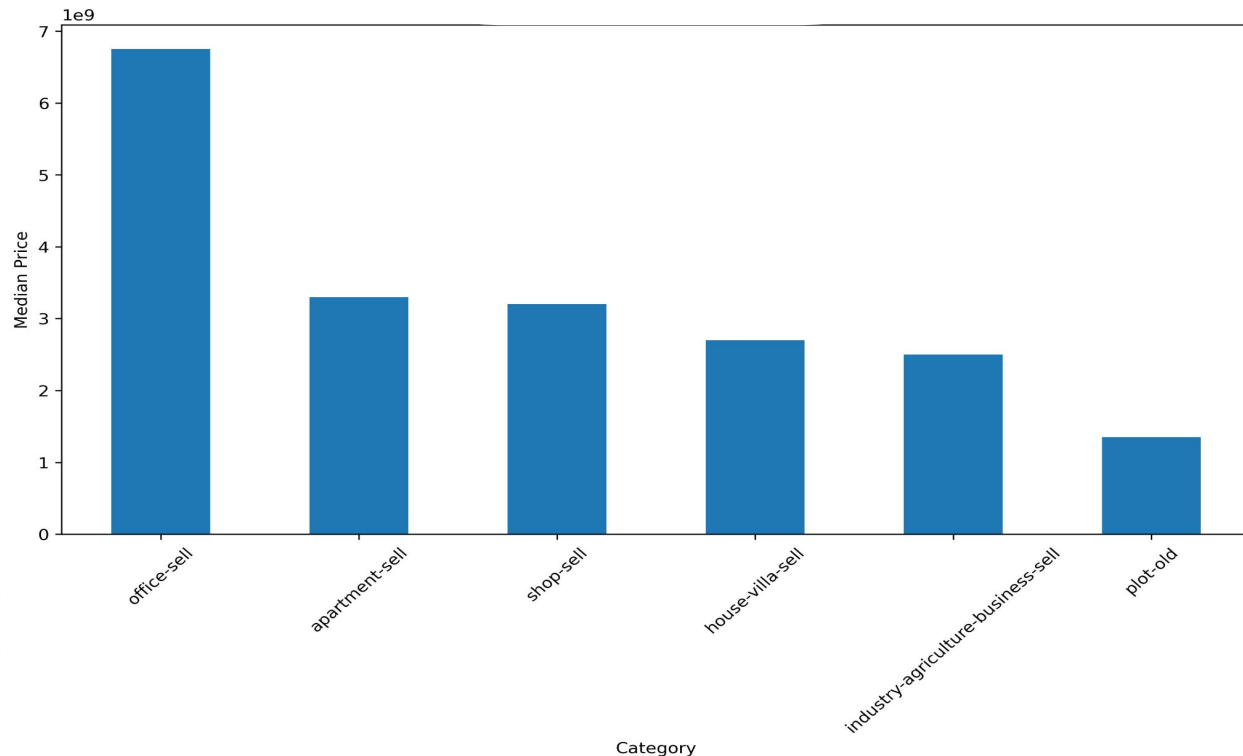




Distribution of Sell Ads by Median Price

Conclusion

- Office listings have the highest median price.
- Apartment and shop properties have similar median prices.
- Plots have the lowest median price among the listed categories.





Research Question:

The average residential house area in metropolitan cities is smaller than in small cities and rural areas.

Conclusion

Because the data distribution was not normal, the Mann-Whitney U test was used. Based on

- the results, the null hypothesis (H0) is rejected
- and the the question is based on the alternative hypothesis.

Test Result :

Utest =

83135225091.5

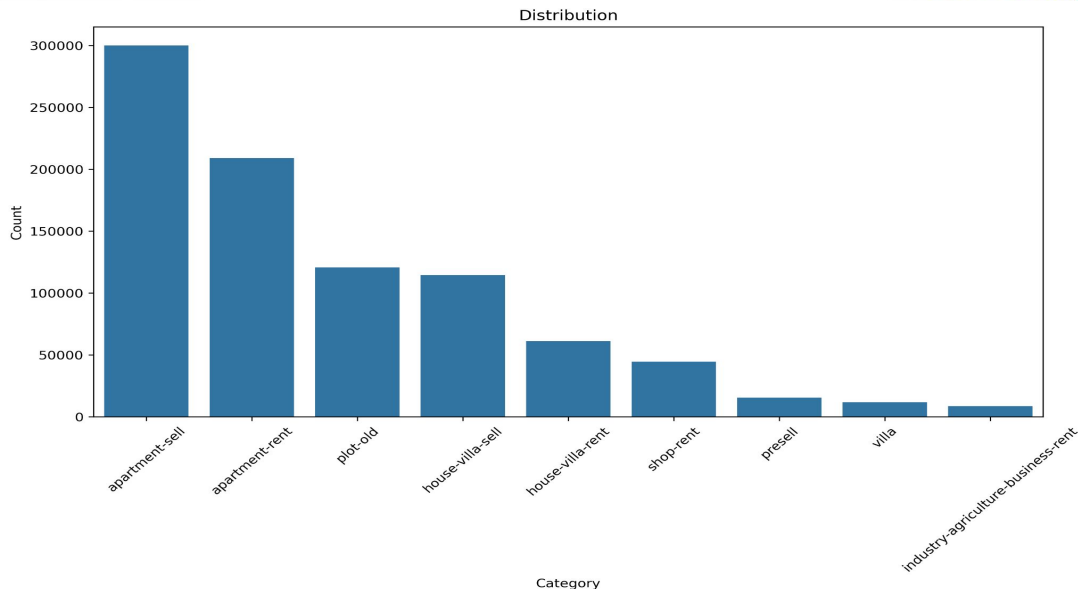
p= 0.0000000045

If $p < 0.05$: reject

H0

Else:

Fail to reject H0



H0: MertroCitiesHouseArea >= Small/RuralCitiesHouseArea

H1: MertroCitiesHouseArea < Small/RuralCitiesHouseArea





Features Used for sell Price Prediction

19 Input Features After Data Cleaning & Feature Engineering



Property Characteristics



- Building Size



- Rooms Count



- Balcony



- Elevator



- Parking



- Warehouse



Location & Time Information



- City



- Property Category (cat2_slug)



- Property Type (cat3_slug)



- Location Latitude



- Location Longitude



- Created Month



- Created Year



Building Information



- Building Age



- Floor



- Total Floors Count



- Units per Floor (Unit_per_floor)



Target Variable



Price Value

The model predicts the price value amount of sell properties.



These 19 features were carefully selected based on domain knowledge, data availability, and exploratory analysis to maximize the predictive power of the model.





SELL PRICE PREDICTION

Model Development & Comparison



Our Modeling Workflow



1. Data Cleaning

Handled missing values, outliers, and inconsistent data.



2. Feature Engineering

Created new features and encoded categorical variables.



3. Bayesian Optimization

Optimized hyperparameters for each model using Bayesian search.



4. Model Evaluation

Compared models using multiple evaluation metrics.



Model Comparison

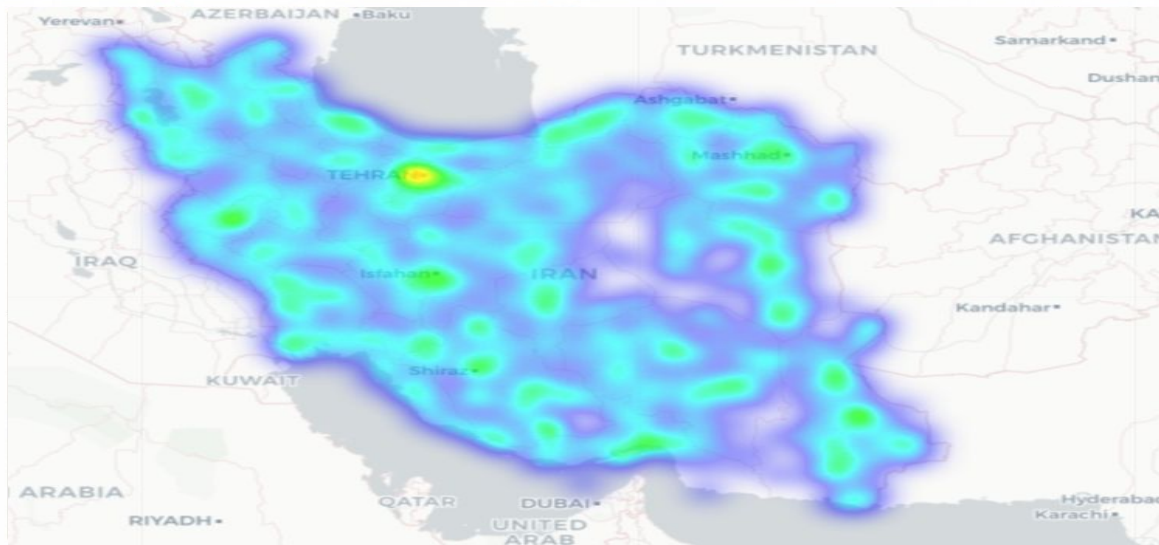
Model	MAE (Million)	RMSE (Million)	R ²
1. Linear Regression	0.305	0.481	0.504
<u>2. Catboost</u>	0.210	0.400	0.656
<u>3. XGBoost (FINAL)</u>	0.205	0.396	0.664





Heatmap of Real Estate Listings

Which regions have the highest concentration of real estate listings in the Divar dataset?



Method

- GeoJSON boundaries
- Valid coordinates
- Folium HeatMap
- Sorted by value counts of city_slug



Key Findings

- **Tehran** has the highest concentration of real estate listings.
- Other major cities, including **Mashhad**, **Karaj**, **Isfahan** and **Shiraz** also show high listing densities.
- Rural and sparsely populated regions contain significantly fewer listings.
- The distribution of listings is **not uniform** across the country and is **strongly** concentrated in large metropolitan areas.



Conclusion

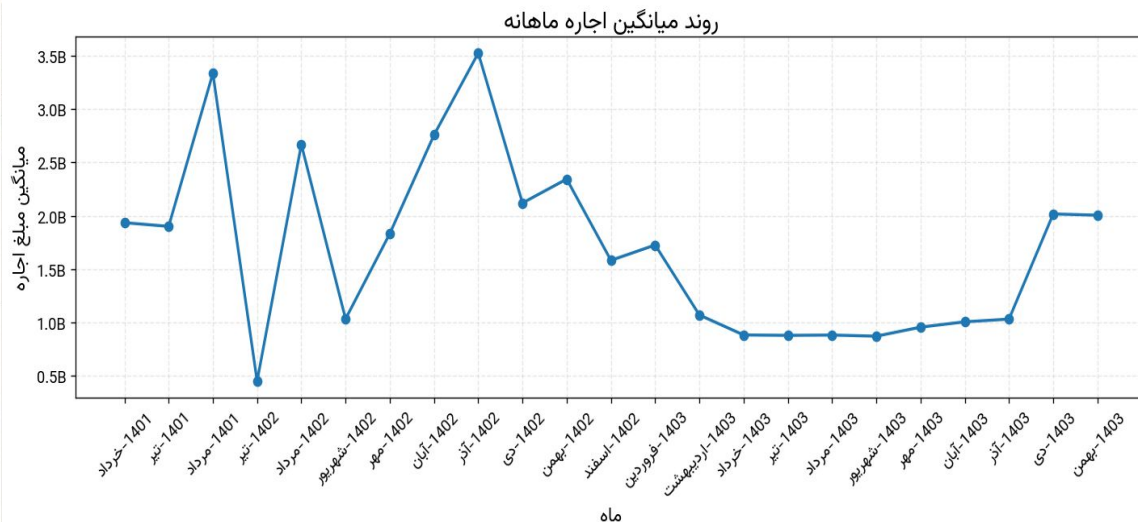
The **heatmap** reveals that real estate advertisements are heavily clustered urban centers, particularly in **Tehran**. This pattern likely reflects higher population density, greater housing market activity and increased usage of the Divar platform in large cities compared to rural areas.





Trend of Average Rental Price by Listing Month

How has the average rental price changed across the Jalali months in which the listings were published?



Method

- Filtered rental listings.
- Converted rent and deposit to **Full Deposit**.
- Removed the top 0.5% outliers (Q99.5).
- Calculated the monthly average by **Jalali month**.



Key Findings

- Rental prices fluctuate across months.
- The highest average occurred in **Azar 1402**.
- Shows a more stable trend **1403**.



Conclusion

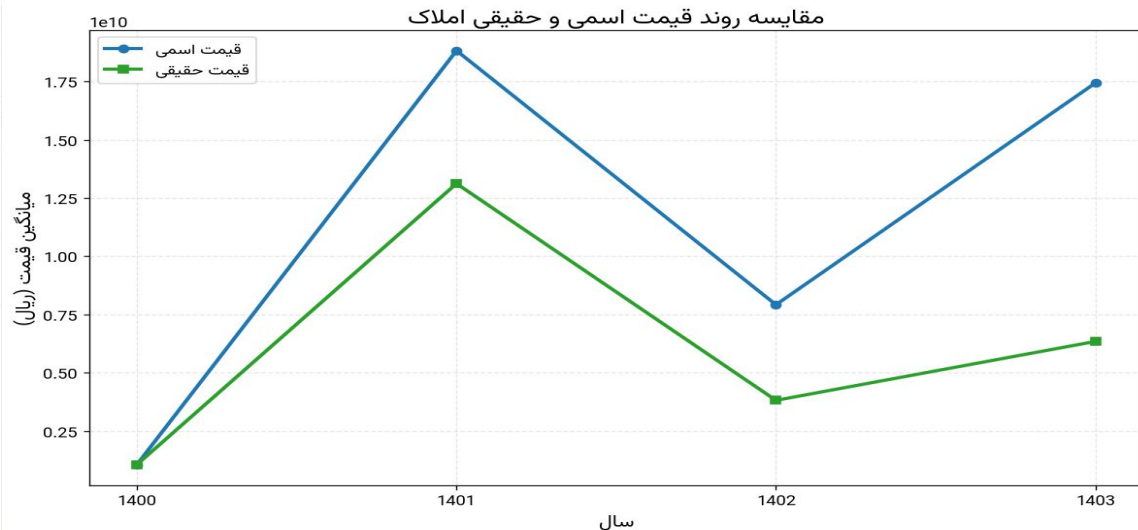
The monthly average **full deposit** over time, indicating that rental prices are influenced by **seasonal** and **market** conditions rather than a consistent **long-term trend**.





Trend of Average Rental Price by Listing Month

How has the average rental price changed across the Jalali months in which the listings were published?



Method

- Calculated the average **nominal price** for each year (1400–1403).
- Adjusted prices using the **Consumer Price Index (CPI)** (base year: 1400).
- Compared nominal and inflation-adjusted (real) price trends.



Key Findings

- Nominal prices increased over time.
- Real prices grew more slowly after adjusting for inflation.
- The gap between nominal and real prices widened in later years.



Conclusion

Although nominal property prices increased from 1400 to 1403, much of this growth was driven by inflation. After inflation adjustment, the real price trend was noticeably more moderate, indicating that the increase in property values was smaller than suggested by nominal prices alone.



Hypothesis Test: Are Old Houses Larger Than New Houses?

Testing the claim: "Old houses were more spacious!"



RESEARCH QUESTION

Are houses built before 1396 (old) larger on average than houses built from 1396 and later (new)?



Old (<1396)

VS



New (≥1396)



HYPOTHESES

$H_0: \mu_{old} \leq \mu_{new}$ (Old houses are not larger)

$H_1: \mu_{old} > \mu_{new}$ (Old houses are larger)



WHY WELCH'S t-TEST?



Very large sample size (>400,00 in each group)



Variances between the two groups are unequal



Welch's t-test is more robust and reliable under these conditions



t-statistic
-148.36 or -16.97

TEST RESULTS

p-value
1.0000

Decision ($\alpha = 0.05$)

Fail to Reject H_0



CONCLUSION

There is **no statistical evidence** to support the claim that old houses are larger than new houses. In fact, **new houses are slightly larger** on average.





Features Used for Rental Price Prediction

19 Input Features After Data Cleaning & Feature Engineering



Property Characteristics



- Building Size



- Rooms Count



- Balcony



- Elevator



- Parking



- Warehouse



Location & Time Information



- City



- Property Category (cat2_slug)



- Property Type (cat3_slug)



- Location Latitude



- Location Longitude



- Created Month



- Created Year



Building Information



- Building Age



- Floor



- Total Floors Count



- Units per Floor (Unit_per_floor)



Target Variable



Full Deposit (Rental Value)

The model predicts the full deposit amount of rental properties.



These 19 features were carefully selected based on domain knowledge, data availability, and exploratory analysis to maximize the predictive power of the model.





Rental Deposit Prediction

Model Development & Comparison



Our Modeling Workflow



1. Data Cleaning

Handled missing values, outliers, and inconsistent data.



2. Feature Engineering

Created new features and encoded categorical variables.



3. Bayesian Optimization

Optimized hyperparameters for each model using Bayesian search.



4. Model Evaluation

Compared models using multiple evaluation metrics.



Model Comparison

Model	MAE (Million)	RMSE (Million)	R ²	Adjusted R ²
1. Linear Regression	294.18	440.51	0.4581	0.4546
2. Ridge Regression	294.75	441.01	0.4569	0.4533
3. Random Forest + Bayesian Optimization	191.67	326.88	0.7016	0.6997
★ 4. XGBoost + Bayesian Optimization (Final Model)	189.86	324.20	0.7069	0.7055



Tree-based ensemble models clearly outperformed linear regression models.
The best performance is achieved by **XGBoost with Bayesian Optimization**.





Final Model Performance

XGBoost + Bayesian Optimization (Best Model)



Best Model: XGBoost + Bayesian Optimization

R^2
0.7069

Adjusted R^2
0.7055

MAE
189.86 M
(Million Toman)

RMSE
324.20 M
(Million Toman)



Validation Set Performance

Metric	Value	Interpretation
MAE	189.92 M	Average absolute error is about 189.92 million.
RMSE	324.92 M	Typical prediction error is about 324.92 million.
R^2	0.7052	The model explains 70.52% of the variance.
Adj. R^2 Adjusted R^2	0.7031	Adjusted for complexity, explains 70.31% of variance.



Very Close
Results



Test Set Performance (Final Evaluation)

Metric	Value	Interpretation
MAE	189.86 M	Average absolute error is about 189.86 million.
RMSE	324.20 M	Typical prediction error is about 324.20 million.
R^2	0.7069	The model explains 70.69% of the variance.
Adj. R^2 Adjusted R^2	0.7055	Adjusted for complexity, explains 70.55% of variance.



Conclusion

- ✓ The validation and test results are **almost identical** across all metrics.
- ✓ This indicates **strong generalization** ability and minimal overfitting.
- ✓ The model can be **reliably used** for predicting rental deposit values.





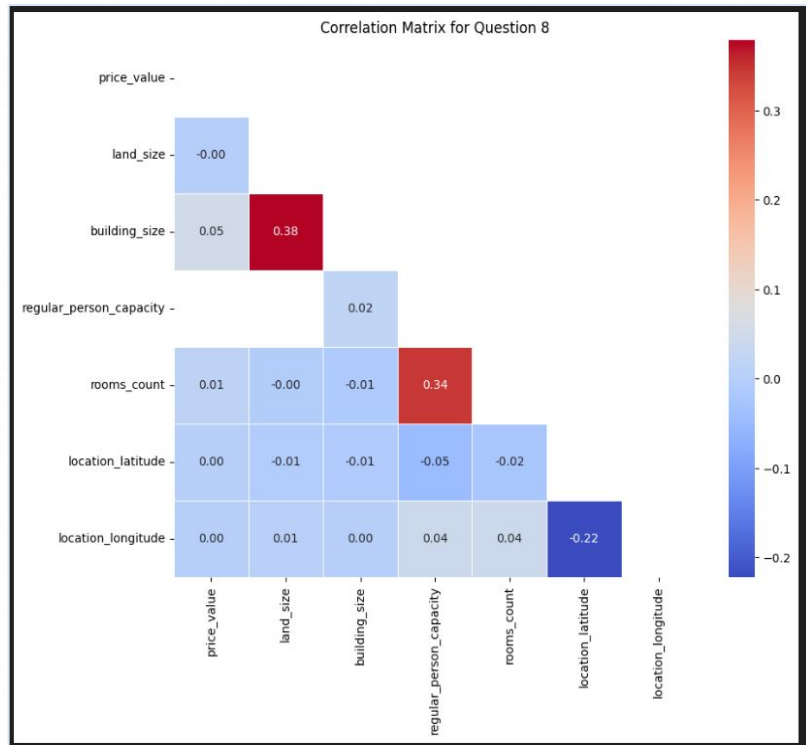
Correlation Matrix

Approach

- Mapped rooms_count from Persian labels to numbers.
- Removed invalid non-positive values for price and area fields.
- Calculated Pearson correlation and visualized it as a masked heatmap.
- Some cells in the correlation matrix are empty because there are no rows where both corresponding columns have no non_null values.

Conclusion

- Most correlations are close to zero, so the variables do not show strong linear dependency.
- The strongest positive pairs are land_size vs. building_size (0.38) and capacity vs. rooms_count (0.34).
- The correlation between latitude and longitude (-0.22) does not mean that one causes the other to change. It only shows the location pattern of the real estate data. Because many listings are from specific cities and areas, these two features have a weak correlation





Distribution of Amenities in Iran

Heatmap analysis of listings with balcony, elevator, security guard, barbecue, and pool



Objective

Identify the main regions where properties with premium amenities are concentrated.



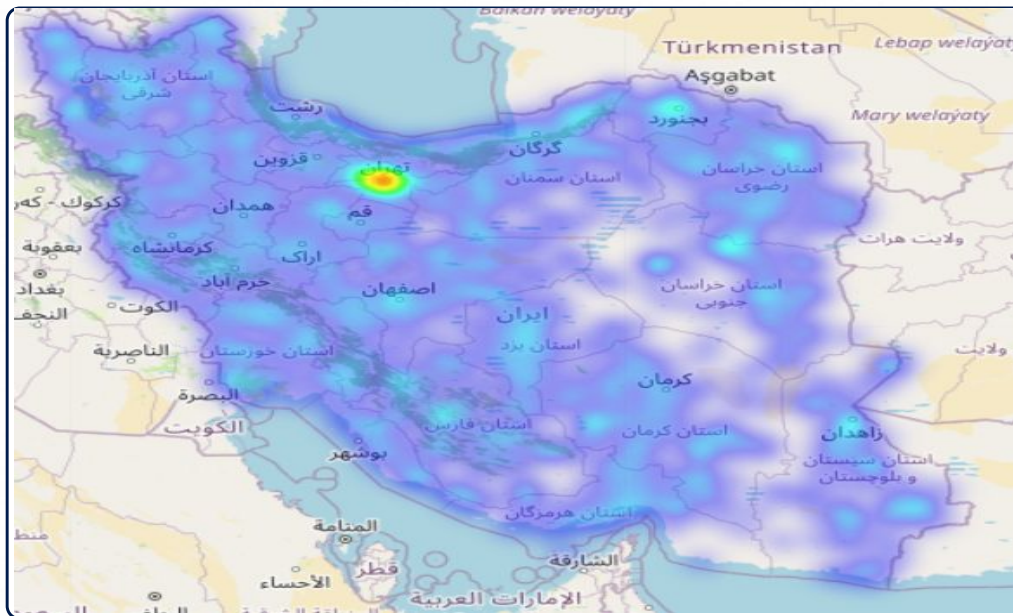
Approach & Tools

- Selected five amenities and location features
- Mapped True/False/true/false values to 1/0
- Handled unknown and missing values
- Kept listings with selected amenities by .all(axis=1)
- Removed invalid geographic coordinates
- Used geopandas to keep points inside Iran
- Created a heatmap with Folium



Conclusion

The strongest hotspot appears around Tehran. Properties with the selected amenities are not evenly distributed across Iran. The main concentration appears around Tehran.



Next step: focus on Tehran neighborhoods for a more detailed analysis.





Tehran Neighborhood Analysis

Top Tehran neighborhoods with the highest number of premium-amenity listings



Why Tehran?

The national heatmap showed that Tehran is the main hotspot, so we continued with a deeper neighborhood-level analysis.



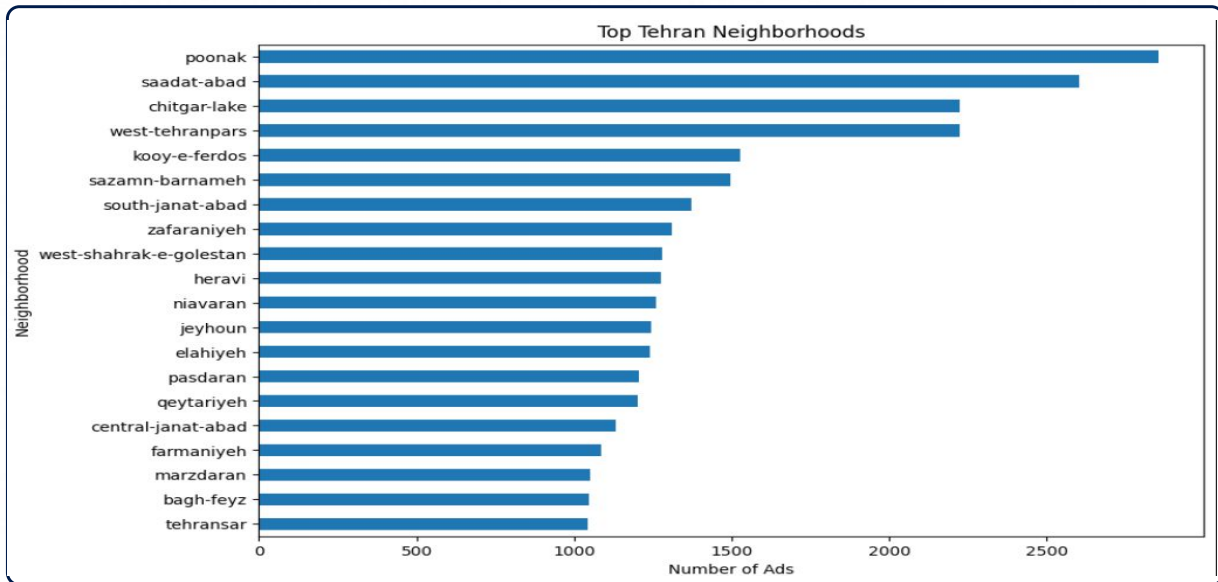
Approach & Tools

- Filtered the data to keep only Tehran listings
- Added city_slug and neighborhood_slug information
- Used Pandas value_counts() to rank neighborhoods
- Used Matplotlib to visualize the top 20 neighborhoods



Conclusion

Within Tehran, premium-feature listings are concentrated in a limited group of neighborhoods. Poonak, Saadat-Abad, Chitgar Lake, and West Tehranpars stand out as the main hubs.





Hypothesis Test 3 – Business Deed and Commercial Sale Price

Testing whether business deed status is related to the sale price of commercial properties

?

Research Question

Does having a business deed make a significant difference in the sale price of commercial properties?

H_0

Null Hypothesis

$H_0: \mu_{\text{with deed}} = \mu_{\text{without deed}}$

There is no significant difference in the mean sale price.

H_1

Alternative Hypothesis

$H_1: \mu_{\text{with deed}} \neq \mu_{\text{without deed}}$

There is a significant difference in the mean sale price.

α

Decision Rule

$\alpha = 0.05$

If $p\text{-value} < 0.05 \rightarrow \text{reject } H_0$

If $p\text{-value} \geq 0.05 \rightarrow \text{fail to reject } H_0$

Variables from Notebook

cat2_slug = commercial-sell

Target variable

log_price

Original price field

price_value

Group field

has_business_deed

Notebook Logic

```
df_hyp3 = df_hyp3[df_hyp3['cat2_slug'] == "commercial-sell"].copy()
price_without_deed = df_hyp3[df_hyp3["has_business_deed"] == False]["log_price"]
price_with_deed = df_hyp3[df_hyp3["has_business_deed"] == True]["log_price"]
```

Why this setup?



Commercial focus

Only commercial sale listings are used, because the hypothesis is about commercial property price.



Independent groups

A property is either with a business deed or without it. The two groups are not paired.

Goal: check if the observed price difference is statistically meaningful, not just random noise.





Hypothesis Test 3 – Preprocessing and Test Selection

From raw commercial listings to a reliable non-parametric hypothesis test

Preprocessing Steps

- 1 Remove missing business deed values
- 2 Remove missing or invalid prices
- 3 Handling outliers
- 4 Histogram
- 5 Split data with deed / without-deed groups
- 6 Check normality with Shapiro test

Normality Check

Shapiro test was applied to both log-price groups.

Result: both groups did NOT look normally distributed.

So an independent t-test was not the best choice.

Chosen Test

Mann-Whitney U test

Why? It compares two independent groups without assuming normal distribution.

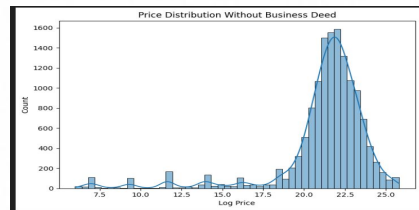
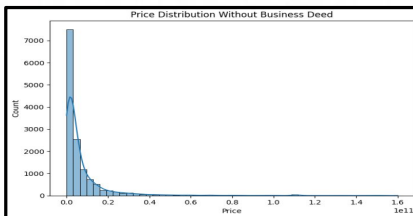
two-sided, because H_1 asks for any difference, not only higher price.

Why Outlier Handling and Log Transform?

Commercial prices had a very large range. Some values were extremely high or low. We delete the 1st and 99th percentiles to reduce extreme effects.

as you can see in the histogram plots, the range between the minimum and maximum values is still large. To improve this problem, we use the log of the price value column.

```
df_hyp3["log_price"] = np.log1p(df_hyp3["price value"])
```



Test Selection Logic

Not normal



Independent groups



Mann-Whitney

This choice makes the analysis more reliable for skewed real-estate prices.





Hypothesis Test 3 – Result and Interpretation

Mann–Whitney U test on log sale prices for commercial properties

Test Report

Test used

Mann–Whitney U

Alternative

two-sided

Alpha

0.05

p-value

< 0.05

Decision

Reject H_0

Meaning

The difference between the two groups is statistically significant in this dataset.

Final Statistical Conclusion

Because the p-value is smaller than 0.05, we reject the null hypothesis.

This means there is a significant difference in log sale prices between commercial properties with a business deed and without a business deed.

Business Interpretation

Business deed can be an important market feature. It may reflect legal clarity and commercial usability, so buyers may value these properties differently.





Hypothesis Test 4 – Luxury Amenities and Sale Price

Testing whether luxury amenities are related to property sale price

?

Research Question

Do listings with luxury amenities have a significant difference in sale price compared with non-luxury listings?

H_0

Null Hypothesis

$H_0: \mu_{\text{luxury}} = \mu_{\text{not luxury}}$

There is no significant difference in mean sale price.

H_1

Alternative Hypothesis

$H_1: \mu_{\text{luxury}} \neq \mu_{\text{not luxury}}$

There is a significant difference in mean sale price.

α

Decision Rule

$\alpha = 0.05$

If p-value < 0.05 → reject H_0

If p-value ≥ 0.05 → fail to reject H_0

Variables from Notebook

Target variable

log_price

Original price field

price_value

Group field

has_luxury
Group flag

cat2_slug = commercial-sell

Luxury Amenities Used

pool

barbecue

jacuzzi

sauna

has_luxury

Notebook Logic

```
df_hyp4["has_luxury"] = (df_hyp4[luxury_cols].fillna(False).any(axis=1))  
  
luxury_ads = df_hyp4[df_hyp4["has_luxury"] == True].copy()  
  
non_luxury_ads = df_hyp4[df_hyp4["has_luxury"] == False].copy()
```



Hypothesis Test 4 – Preprocessing and Test Selection

From raw real-estate listings to a reliable non-parametric hypothesis test

Preprocessing Steps

1 Select luxury columns and price_value

2 Keep only valid positive prices

3 Remove rows with unknown luxury status

4 Create has_luxury flag

5 Split data into luxury / non-luxury groups

6 Trim bottom 2% and top 2% prices by group

7 Create $\log_price = \log(p(\text{price_value}))$

8 Check normality using Shapiro test

9 Apply Mann-Whitney U test

Why Outlier Handling and Log Transform?

Real-estate prices have a very large range. Some values are extremely high or low. We removed the bottom 2% and top 2% of prices inside each group, then used $\log(p(\text{price_value}))$ to make the distribution less skewed and easier to compare.

Normality Check

Shapiro test was applied to both log-price groups.

Result: both groups did NOT look normally distributed.

So a classic independent t-test was not the safest choice.

Chosen Test

Mann-Whitney U test

Why? It compares two independent groups without assuming normal distribution.

two-sided, because H_1 asks for any difference.

Test Selection Logic

Not normal

Independent groups

Mann-Whitney

This choice makes the analysis more reliable for skewed real-estate price data.





Hypothesis Test 4 – Result and Interpretation

Mann–Whitney U test on log sale prices for luxury vs non-luxury ads

Test Report

Test used
Mann–Whitney U

Alternative
two-sided

Alpha
0.05

p-value
< 0.05

Decision

Reject H_0

Meaning

The difference between the two groups is statistically significant in this dataset.

Final Statistical Conclusion

Because the p-value is smaller than 0.05, we reject the null hypothesis. This means there is a significant difference in log sale prices between ads with luxury amenities and ads without luxury amenities.

Business Interpretation

Luxury amenities can act as important price signals. Listings with pool, barbecue, jacuzzi, or sauna usually belong to higher-value property segments.





Recommender System – KMeans Clustering Strategy

Goal: create meaningful property groups based on user needs,(price level, and location are the most important features)



Main Objective

Cluster real-estate ads so a recommender can show similar properties to a user instead of searching the full dataset.



Why Feature Selection Matters

Clusters must describe user preferences. We used a small number of strong features to avoid noisy clusters and the curse of dimensionality.



KMeans Setup

First, KMeans was run with $k = 10$ for the required visual comparison. Then (part2 k was tuned from 1 to 20.

Feature sets tested

Set 1

price_value + latitude + longitude

baseline: sale price + location

Set 2

**price_value + latitude + longitude
+ rooms_count + building_size**

more user detail, but more missing values

Set 3

fullll_price + latitude + longitude

covers both sale and rent/credit ads

How fullll_price was created

```
full_credit = rent/30000*1000000  
+ credit
```

For rental ads, rent and credit were converted into one comparable credit-like value. Then fullll_price used price_value when available, otherwise full_credit. This made the clustering more complete.

Final selected feature idea

```
final_price + UTM/location
```





Recommender System – Preprocessing Pipeline

Main preprocessing steps

- 1 Select core fields**
price, rent, credit, location, rooms, building size
- 2 Map rooms_count**
Persian labels → numeric values: 0 to 5
- 3 Create final_price**
combine sale and rent/credit advertisements
- 4 Validate geography**
use Iran GeoJSON + GeoPandas
- 5 Remove invalid rows**
missing values, zero/negative prices, impossible sizes
- 6 Trim outliers**
2% to 98% quantile filtering for price fields
- 7 Scale features**
StandardScaler before KMeans
- 8 Evaluate clusters**
inertia, silhouette, Calinski-Harabasz, Davies-Bouldin

Why scaling?

KMeans uses distances. Without scaling, price could dominate location or size only because its numeric range is bigger.

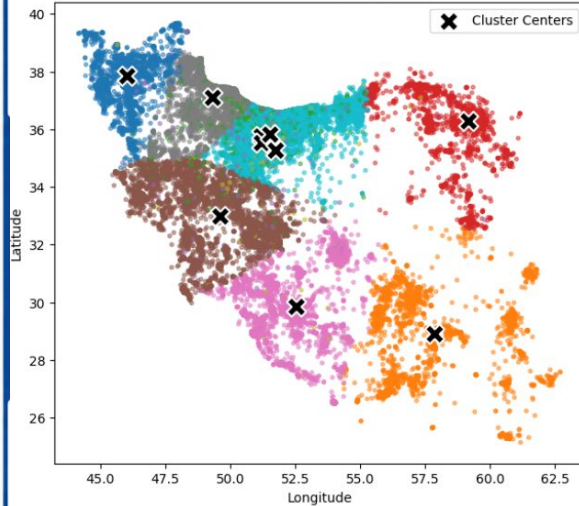
	feature_set	inertia	silhouette_score	calinski_harabasz_score	davies_bouldin_score
0	price_location	156170.233158	0.420288	230607.584158	0.828187
1	price_price_loc_room_size	424456.804299	0.333715	69497.704071	1.106740
2	final_price_loc	245849.236362	0.452302	391998.129789	0.869849



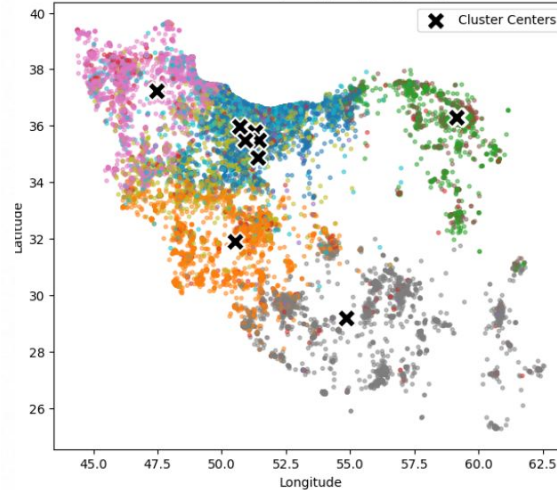


KMeans Part 1 – Feature Set Comparison

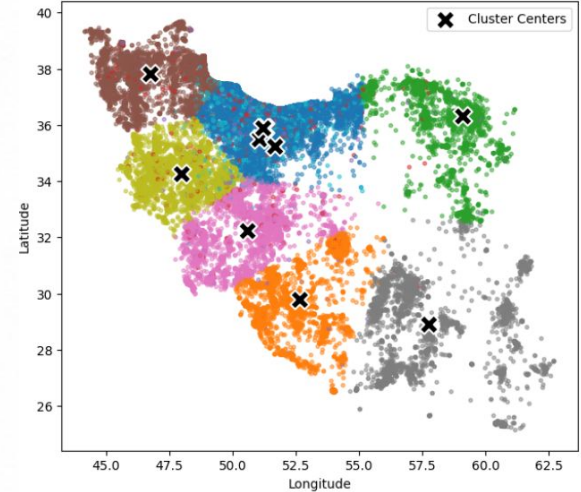
KMeans Clusters - price_location



KMeans Clusters - price_price_loc_room_size



KMeans Clusters - final_price_loc



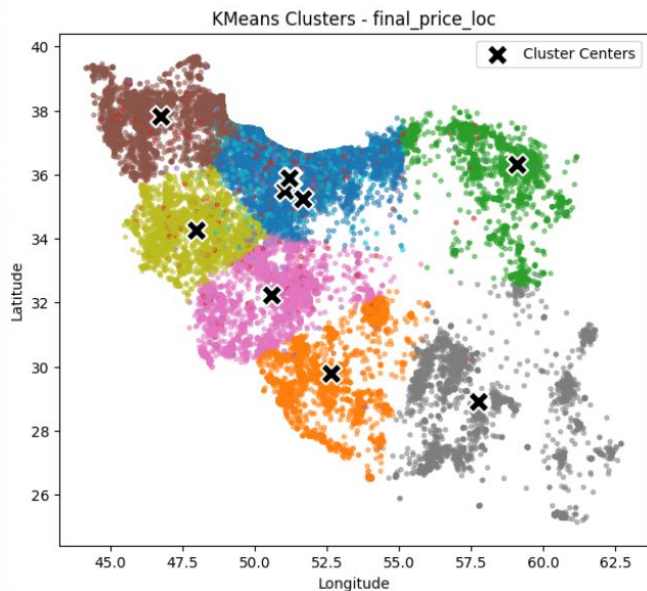
Based on the scores and the visual results, I choose the (final_price + location). This feature set had better scores, especially a higher silhouette score and a higher Calinski-Harabasz score. In the visualization, the clusters are also clearer and more meaningful. Also, this plot includes both sale ads and rent ads. Because of this, it gives a more complete view of the real estate data. So, I think this feature set is the better choice.





KMeans Part 1 – UTM Conversion and Final 10-Cluster View

After choosing final_price + location, geographic coordinates were converted to UTM for metric plotting



Selected final_price_loc cluster layout; notebook also converts lat/lon to UTM X/Y

Why UTM?



Latitude and longitude are in degrees. UTM converts them to meter-based X/Y coordinates, which are easier to interpret for distance and spatial plots.

Conversion logic from notebook

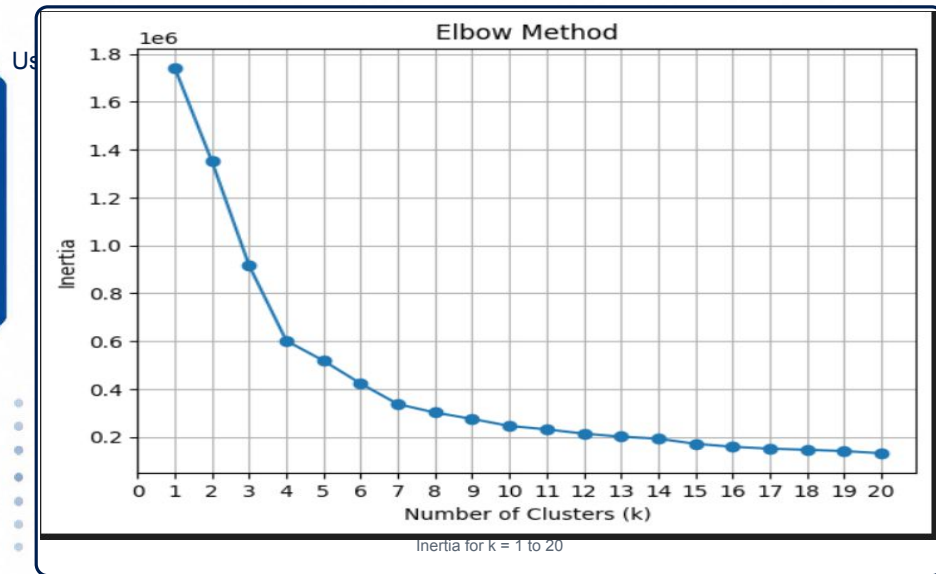
```
utm.from_latlon(lat, lon, force_zone_number=39)
```

The same cluster labels and center points were transformed into UTM coordinates. The plot then uses UTM X and UTM Y instead of longitude and latitude.





KMeans Part 2 – Elbow Method



Σ

What inertia means

Inertia is the within-cluster sum of squared distances. Lower inertia means points are closer to their assigned center.



What we observed

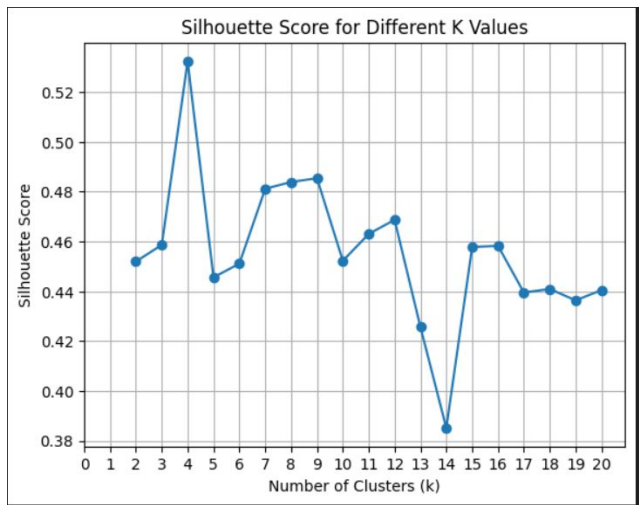
The curve drops fast until around $k = 4$. There is still a visible improvement until around $k = 7$, but after that the curve becomes flatter.





KMeans Part 2 – Silhouette Score

To make a more reliable decision, we also compare the silhouette scores.



Silhouette score for k = 2 to 20

What silhouette means

Silhouette score checks how well each point fits its own cluster compared with other clusters. Higher is better.

Observed result

The highest score appears at $k = 4$. Scores around $k = 7$ to $k = 9$ are also relatively strong and more detailed.

Why $k = 7$?

from 4 to 7 clusters, there is still a clear decrease in inertia. Also, in my opinion, dividing all real estate data in Iran into only 4 groups is not logical because Iran has a large geographical area. Therefore, 7 clusters seem to be a better choice.





Recommender System – DBSCAN Clustering

Density-based clustering using UTM coordinates and comparable property price



Goal

Create 3 meaningful clusters for real-estate listings, based on dense spatial-price patterns.



Inputs

- UTM X
- UTM Y
- Comparable price

Main Idea

DBSCAN groups points that are close to each other in scaled UTM-price space. Points that do not belong to any dense area are treated as noise.

1

Choose features

UTM X/Y + price

2

Scale data

StandardScaler

3

Tune eps

neighborhood radius

4

Tune min_samples

density threshold

5

Check result

clusters + noise

Hyperparameter Effects

- larger eps → points connect more easily
- larger min_samples → denser clusters needed
- both control number of clusters and noise

Final Target

- 3 clusters, as required by the project
- keep enough noise control
- make clusters interpretable on the map

Key point: DBSCAN does not force every listing into a cluster; noisy or isolated listings can stay outside the main groups.





DBSCAN – Preprocessing Pipeline

From raw Divar listings to scaled UTM-price features

Comparable Price Logic

```
full_credit = rent_value / 30,000 × 1,000,000 + credit_value  
price_value = price_value.fillna(full_credit)
```

Cleaning and Feature Engineering Steps

- Selected rent_value, price_value, credit_value, latitude, longitude
- Created full_credit to include rent/credit listings
- Filled missing sale price using full_credit
- Kept only positive valid prices
- Trimmed extreme prices using 2%–98% quantiles
- Filtered coordinates to Iran bounding box
- Removed rows with missing latitude or longitude
- Used Iran GeoJSON + GeoPandas spatial join
- Converted lat/lon to UTM X and UTM Y (zone 39)
- Scaled UTM X, UTM Y, and price using StandardScaler

Why these steps matter



Price conversion

Sale and rent/credit ads become comparable in one feature.



Outlier trimming

Extreme property prices can distort distance-based clustering.



Geographic validation

Coordinates outside Iran or invalid points are removed.



UTM conversion

Latitude/longitude degrees are converted to meter-based coordinates.



Scaling

DBSCAN uses distance; scaling stops price from dominating location.

Final feature matrix used for DBSCAN: [utm_x, utm_y, comparable price]



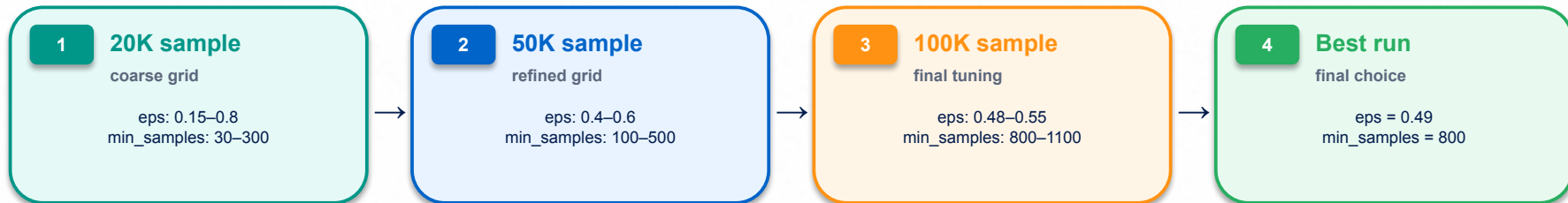


DBSCAN – Sampling and Hyperparameter Search

We started small, found a useful range, then increased sample size for a stronger result

Search strategy

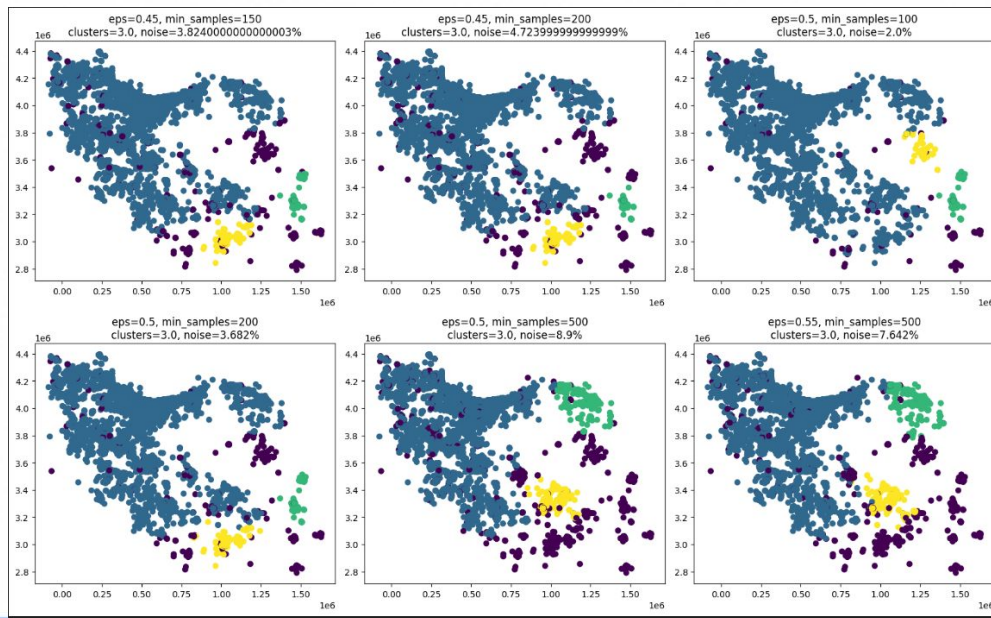
Running DBSCAN on the full dataset caused memory issues, so the process used representative samples and gradually increased their size.





DBSCAN – Tuning Process and Final Result

Visual comparison of candidate parameters and final 3-cluster output



Selected Configuration

sample size = 100,000
 $\text{eps} = 0.49$
 $\text{min_samples} = 800$
number of clusters = 3
noise percent : 8.025

